

Maximum Number of Electron Diffraction Rings

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Abstract

In a model of the electron diffraction experiment by Lester Germer and Clinton Davisson the number of electron diffraction rings can be calculated using Bragg's law and the de Broglie wavelength equation.

Maximum value of n

Where n is the number of rings, h is planks constant, m is mass of electron, e is charge of electron, V is potential difference of $5kV$ and d is the atomic spacing in layers of graphite atoms. The C-C bond length in graphite is approximately 0.142 nm (Sun et al., 2009). Therefore d_1 is $\frac{\sqrt{3}}{2} \times 0.142 \times 10^{-9} = 0.123 \times 10^{-9}$ (3sf) and d_2 is $\frac{3}{2} \times 0.142 \times 10^{-9} = 0.213 \times 10^{-9}$.

n is in the relationship:

$$\sin \frac{\phi}{2} = \frac{nh}{2d\sqrt{2meV}}$$

Isolating n :

$$\frac{\sin \frac{\phi}{2} 2d\sqrt{2meV}}{h} = n$$

Due to the maximum value of $\sin$ being 1:

$$\begin{aligned} \frac{2d\sqrt{2meV}}{h} &\geq n \\ \therefore \frac{2d\sqrt{2meV}}{h} &= n_{\max} \end{aligned}$$

When d is equal to 0.123×10^{-9} : $n_{\max} = 14.2$ (3sf)

When d is equal to 0.213×10^{-9} : $n_{\max} = 24.5$ (3sf)

As n is the maximum number of rings, the values should be truncated to integers.

$$d = 0.123 \times 10^{-9}, n = 14$$

$$d = 0.213 \times 10^{-9}, n = 24$$